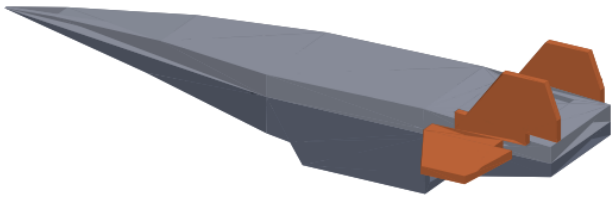
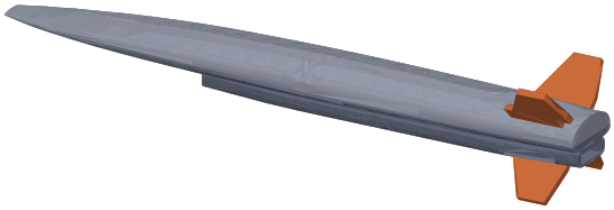
Hypersonic Vehicle Display Models

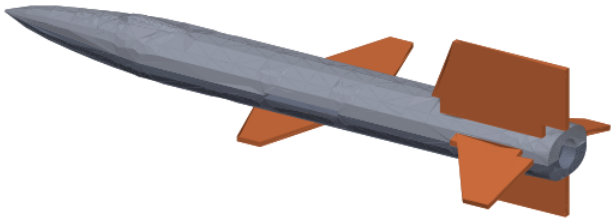
Print-and-assemble kit for PLA+ | X-43A · X-51A · HTV-2 · X-15 · Caret waverider



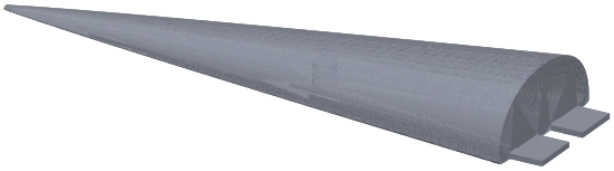
X-43A Hyper-X, 1:24



X-51A Waverider, 1:24



X-15, 1:72



HTV-2, 1:24

About these models. The outer shapes are **approximations** built from published overall dimensions and public photographs. Detailed outer mold lines for these vehicles are not public, so the models are meant for display and teaching, not for wind-tunnel or CFD-grade geometry. The one exception is the caret waverider, which is the exact inviscid shape from oblique-shock theory for its design point.

What's in the kit

Model	Scale	Length	Parts	Real vehicle
X-43A Hyper-X	1:24	153 mm	2 body + 4 surfaces	3.66 m, H2 scramjet, Mach 9.6 (2004)
X-51A Waverider (cruiser)	1:24	178 mm	2 body + 4 fins	4.27 m cruiser, JP-7 scramjet, Mach 5.1 (2013)
HTV-2 Falcon	1:24	163 mm	1 body	~3.7 m boost-glide vehicle (2010, 2011)
North American X-15	1:72	215 mm	2 body + 6 surfaces	15.45 m, rocket plane, Mach 6.70 (1967)
Caret waverider	-	160 mm	1 body	Nonweiler caret, M = 6, theta = 7 deg

Every model also has a display stand (base with engraved label + post). Each part comes as an **STL** (for your slicer) and a **STEP** (opens as an editable solid in Fusion or SolidWorks). The Python source in `src/` regenerates everything and holds all dimensions as parameters.

Printer bed

All parts except the X-15 body halves fit within 180 x 180 mm. The X-15 halves are 215 mm long: they fit a 220 mm (or larger) bed, or lie diagonally on a 180 mm bed (for example a Bambu A1 mini).

Materials and tools

- PLA+ filament, roughly 250-350 g for the whole set (about 40-90 g per model including its stand).
- **Alignment pins:** a short length of 1.75 mm filament. Cut pins to **6.5 mm**. The holes are 1.95 mm wide and 3.5 mm deep in each half.
- Glue: medium CA (super glue) with accelerator, or 5-minute epoxy for a longer working time on the body seams.
- Hobby knife, flush cutters, needle files, sandpaper (220, 400, 800 grit), small square or set square, rubber bands or small clamps.
- Optional: filler putty or filler primer, spray primer, acrylic or enamel paints, decals.

Print settings (PLA+)

Setting	Body parts	Fins / surfaces	Stand
Nozzle	0.4 mm	0.4 mm	0.4 mm
Layer height	0.12-0.16 mm (finer = smoother curves)	0.12 mm	0.2 mm
Walls / perimeters	3-4	4 (effectively solid)	3
Top / bottom layers	5 / 4	solid	4 / 4
Infill	15-20 % gyroid	100 %	15 %
Nozzle / bed temp	205-220 C / 55-60 C (follow your spool)	same	same
Supports	None (split face down)	None	None
Adhesion	Brim on X-43A lower half, HTV-2, caret	None or 3 mm brim	Brim on post

Why split bodies? Each fuselage is cut in half at a flat horizontal plane. Each half prints on that flat face with no supports, and the curved outer skin builds up as gentle overhangs. Fins print flat on the bed, which makes them stronger and keeps the edges crisp.

General assembly technique (all models)

1. Clean up. Remove brims. Shave the 'elephant foot' lip off every split face with a hobby knife held flat, so the two halves close with no gap. Check that the pin holes are clear (twist a 2 mm drill bit by hand if needed).

2. Make the pins. Cut 6.5 mm lengths of 1.75 mm filament, one per pin hole pair. Glue them into the **lower** half only.

3. Dry fit everything before gluing. Close the halves on the pins, then push every fin tab into its slot. Slots have 0.2 mm clearance per side. If a tab is tight, sand its faces, not the slot. Each fin root is trimmed to the curved body, so it seats flush only the right way round, with the swept leading edge facing the nose.

4. Glue the body. Run a thin bead of CA or epoxy around the split face, keeping 2 mm away from the outer edge so glue doesn't squeeze out. Close on the pins and hold with rubber bands for a few minutes.

5. Fit the surfaces. Put a drop of glue on each tab, insert it, and set its angle with a square while the glue grabs. Mirror-image pairs (left and right wings, for example) are the same part flipped over.

6. Finish. Fill the seam with putty or filler primer, then sand 220 → 400 → 800. Prime, check under strong side light, fill again if needed, then paint.

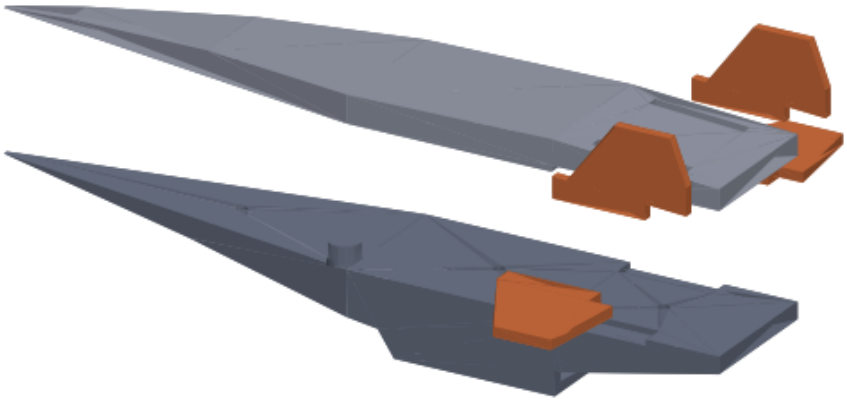
7. Stand. Glue the post into the base. Leave the 6 mm peg unglued in the model's socket so the model lifts off for handling.

Fit tuning: if your printer runs tight, scale **only the fin parts** to 98-99 % in the slicer, or set `CLEAR` in `build_models.py` to 0.25-0.3 mm and regenerate. Don't scale a whole model in the slicer if you want the pins and peg to keep fitting.

1. NASA X-43A Hyper-X (1:24)

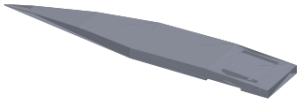
NASA built the X-43A to show that an airframe-integrated, hydrogen-fuelled scramjet could work in flight. It was carried by a B-52B and boosted by a modified Pegasus rocket. On 27 March 2004 it flew at about Mach 6.8, and on 16 November 2004 at about Mach 9.6, a record for an air-breathing vehicle.

What to look for: the whole underside is the engine. The forebody ramps compress the air before the inlet, the cowl and sidewalls form a real flow-through duct (you can see daylight through it), and the upswept aft underside is the single-expansion ramp nozzle (SERN).

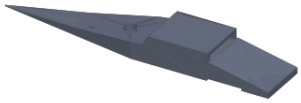


Exploded view. Grey: body halves; orange: printed-flat surfaces.

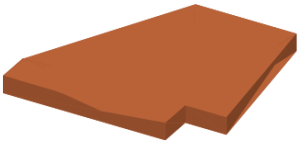
File (stl/ and step/)	Qty	Size mm (as printed)	Print orientation	Supports
x43a_body_upper.stl	1	152 x 28 x 6	Split face down	No
x43a_body_lower.stl	1	152 x 28 x 15	Split face down (cowl on top; the duct roof is a 23 mm bridge). 5 mm brim	No
x43a_wing_x2.stl	2	23 x 22 x 2	Flat	No
x43a_vtail_x2.stl	2	27 x 17 x 2	Flat	No
x43a_stand_base.stl	1	110 x 60 x 5	Flat, label up	No
x43a_stand_post.stl	1	8 x 8 x 76	Standing upright, peg up; 5 mm brim	No



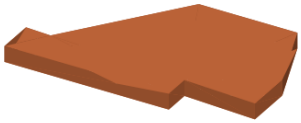
x43a_body_upper



x43a_body_lower



x43a_wing_x2



x43a_vtail_x2

Assembly

1. Glue three pins (6.5 mm) into the lower half. Dry-fit the upper half.
2. Glue the halves (see General technique). The duct lies entirely in the lower half, so keep glue off the cowl.
3. **All-moving wings:** the side slots sit exactly on the seam. Push a wing tab into each side with the long straight root against the body, the swept edge forward, and the wing level (0 deg dihedral).
4. **Vertical tails:** push the tabs into the two slots on the upper aft deck and square them at 90 deg to the deck.
5. Socket: under the forebody ramp, about 69 mm behind the nose.

Paint: refer to NASA Armstrong flight photographs for the scheme. A flat-black overall finish with the underside engine module left slightly lighter makes the flowpath stand out on a display model.

2. X-51A Waverider (1:24, cruiser)

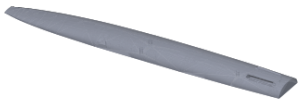
A USAF / DARPA / NASA program with Boeing and Pratt & Whitney Rocketdyne. The SJY61 scramjet burned JP-7, which also cooled the engine. After a B-52H drop and a booster burn, the fourth and final flight on 1 May 2013 reached Mach 5.1, with about 210 s of scramjet operation.

What to look for: the flat, sharp forebody is a waverider surface that pre-compresses air into the underside inlet, so the vehicle rides its own shock. The model has an open inlet and an open nozzle on the ventral nacelle. Only the cruiser is modeled here, not the interstage or booster.

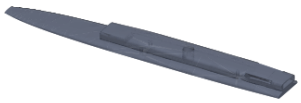


Exploded view.

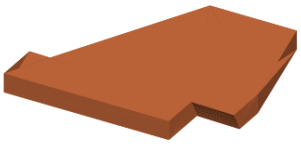
File (stl/ and step/)	Qty	Size mm (as printed)	Print orientation	Supports
x51a_body_upper.stl	1	178 x 23 x 7	Split face down	No
x51a_body_lower.stl	1	178 x 23 x 8	Split face down (nacelle on top)	No
x51a_fin_upper_x2.stl	2	20 x 18 x 2	Flat	No
x51a_fin_lower_x2.stl	2	20 x 18 x 2	Flat (root notched to clear nacelle)	No
x51a_stand_base.stl	1	110 x 60 x 5	Flat, label up	No
x51a_stand_post.stl	1	8 x 8 x 76	Standing upright, peg up; 5 mm brim	No



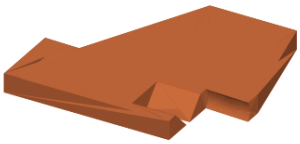
x51a_body_upper



x51a_body_lower



x51a_fin_upper_x2



x51a_fin_lower_x2

Assembly

1. Glue three pins into the lower half: two side by side mid-body and one aft. There are no pins in the thin nose; glue it carefully and clamp the tip.

2. Glue the halves.

3. Fins (X layout, 45 deg): the upper pair goes into the slots in the upper half, the lower pair into the slots beside the nacelle. Keep upper and lower fins apart; their roots are shaped differently. The slot sets the 45 deg angle.

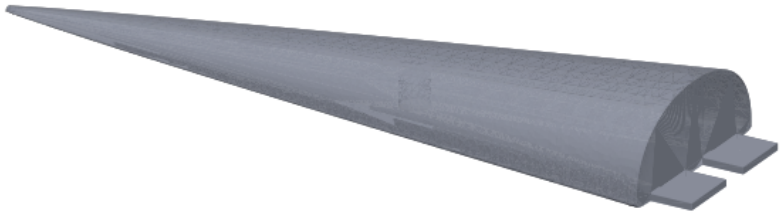
4. Socket: under the nacelle, about 115 mm behind the nose.

Paint: follow USAF / AFRL flight-vehicle photographs.

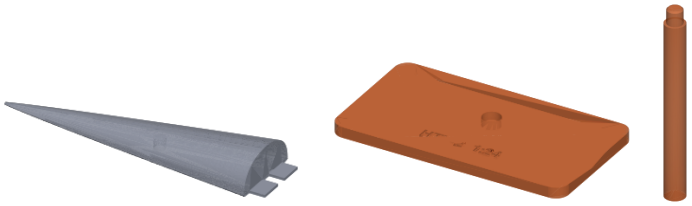
3. DARPA HTV-2 Falcon (1:24)

A hypersonic boost-glide test vehicle built by Lockheed Martin for DARPA's Falcon program. It flew twice on Minotaur IV Lite rockets from Vandenberg, on 22 April 2010 and 11 August 2011, and was meant to glide at about Mach 20 with a carbon-carbon aeroshell. Both flights ended early.

Note: the detailed HTV-2 shape is not public. This model is a generic sharp-nosed, flat-bottomed lifting cone with two body flaps, sized from the published length. Treat it as representative of the class.



File (stl/ and step/)	Qty	Size mm (as printed)	Print orientation	Supports
htv2_body.stl	1	163 x 45 x 19	On its flat bottom; 5 mm brim. Flaps print as part of the first layers	No
htv2_stand_base.stl	1	110 x 60 x 5	Flat, label up	No
htv2_stand_post.stl	1	8 x 8 x 76	Standing upright, peg up; 5 mm brim	No



htv2_body htv2_stand_base htv2_stand_post

Assembly

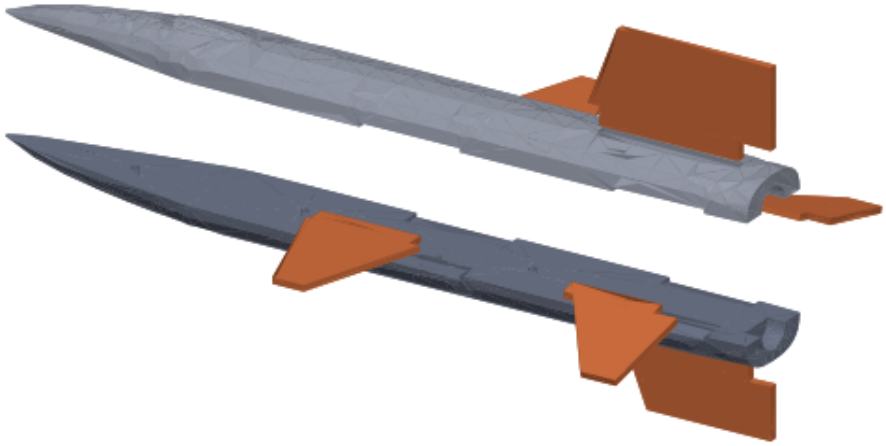
1. Remove the brim carefully around the thin (1.5 mm) body flaps.
2. Sand the upper surface lightly, 400 then 800 grit, along the length.
3. Socket: centred in the flat bottom, about 92 mm behind the nose.

Paint: dark grey / charcoal (carbon-carbon).

4. North American X-15 (1:72)

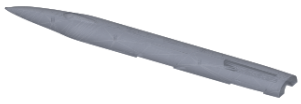
The X-15 was a rocket-powered research aircraft that flew 199 missions between 1959 and 1968 from a B-52 mothership. Pete Knight set the speed record of Mach 6.70 on 3 October 1967 in the lengthened X-15A-2. This model shows the standard-length aircraft.

What to look for: the thick wedge-section vertical tails, which gave stability at high Mach; the side fairings; and the all-moving horizontal tails with 15 deg anhedral. The lower ventral fin was jettisoned before landing, so it's optional on the model.

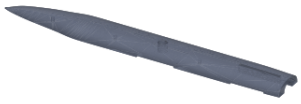


Exploded view.

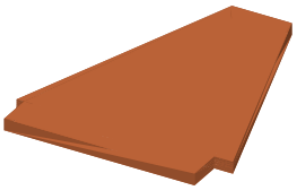
File (stl/ and step/)	Qty	Size mm (as printed)	Print orientation	Supports
x15_body_upper.stl	1	215 x 26 x 10	Split face down	No
x15_body_lower.stl	1	215 x 26 x 9	Split face down	No
x15_wing_x2.stl	2	32 x 39 x 2	Flat	No
x15_htail_x2.stl	2	31 x 30 x 2	Flat	No
x15_upper_vtail.stl	1	51 x 26 x 2	Flat	No
x15_ventral_fin.stl	1	40 x 19 x 2	Flat (optional)	No
x15_stand_base.stl	1	110 x 60 x 5	Flat, label up	No
x15_stand_post.stl	1	8 x 8 x 66	Standing upright, peg up; 5 mm brim	No



x15_body_upper



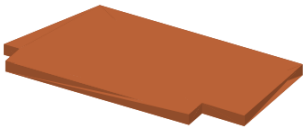
x15_body_lower



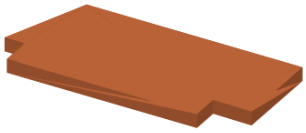
x15_wing_x2



x15_htail_x2



x15_upper_vtail



x15_ventral_fin

Assembly

1. Glue three pins into the lower half, dry-fit, then glue the halves.

2. **Wings:** the side slots are on the seam. Insert both wings level, with the swept edge forward.

3. **Horizontal tails:** the slots are angled, so the tails take up 15 deg anhedral (tips down) on their own. Check that both sides match when viewed from behind.

4. **Upper vertical tail:** insert into the dorsal slot and square it to the wings.

5. **Ventral fin (optional):** insert into the slot under the tail. Leave it off for the post-jettison landing look.

6. Socket: under the fuselage at mid-wing, about 119 mm behind the nose.

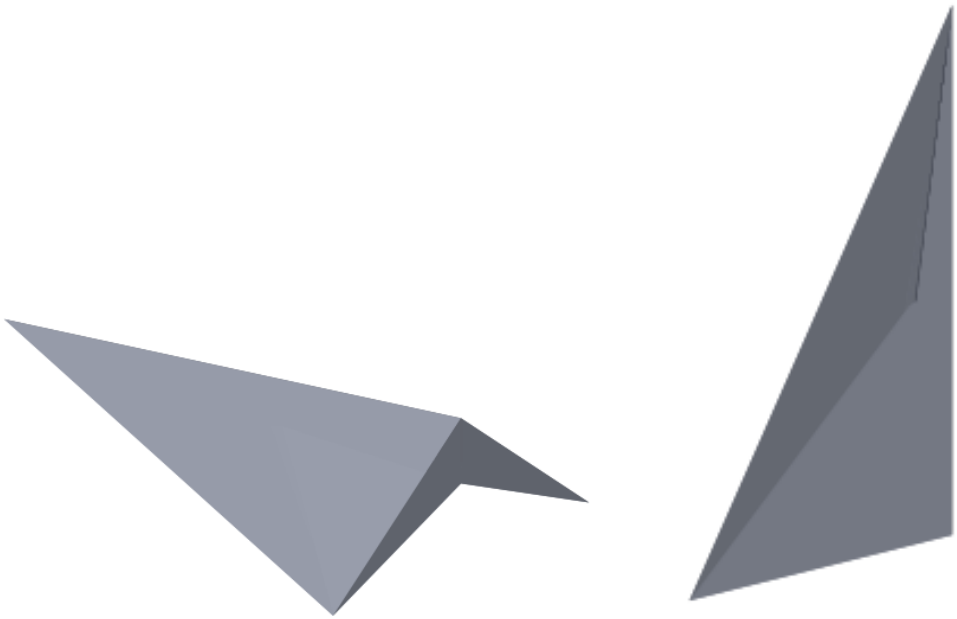
7. Optional: sand a 10 deg wedge onto the vertical tail faces for more accuracy.

Paint: gloss or semi-gloss black overall. This was the heat-resistant black finish on the Inconel X skin.

5. Caret waverider (exact design, M = 6)

Nonweiler's caret waverider is built from a known flow: the 2-D oblique shock from a wedge. The leading edges lie in a single planar shock. The upper surfaces are parallel to the freestream, so they stay at freestream pressure. The two lower surfaces are made of wedge-flow streamlines. At the design point the shock sits exactly on the leading edges, so there's no spillage from the high-pressure underside to the top.

Parameter	Value
Design Mach number	6.0 (gamma = 1.4)
Wedge deflection theta	7.0 deg
Shock angle beta (weak solution)	14.84 deg
Lower-surface pressure ratio p2/p1	2.59 = 1 + 2gamma/(gamma+1) (M^2 sin^2 beta - 1)
Inviscid L/D (pressure only, no base drag)	cot theta = 8.14
Length / span / base height	160 / 128 / 42.4 mm



Left: as displayed. Right: print orientation (standing on its base, nose up).

File (stl/ and step/)	Qty	Size mm (as printed)	Print orientation	Supports
caret_waverider.stl	1	42 x 128 x 160	Standing on its flat base, nose up; 5 mm brim; min layer time 8 s for a clean tip	No
caret_stand_base.stl	1	110 x 60 x 5	Flat, label up	No
caret_stand_post.stl	1	8 x 8 x 76	Standing upright, peg up; 5 mm brim	No

Assembly

1. Print with the brim. The leading edges are knife edges and come out as thin as your nozzle allows. Leave them; don't sand them round.

2. The socket is in the lower ridge at 60 % of the length. Push it onto the stand peg so the flat upper surfaces face up.

3. To make other design points (Mach number, theta, span), edit the `build_caret(L, M, theta_deg, span_ratio)` call and regenerate. The stand label updates to match.

Editing the models in CAD

- **Fusion (free personal license) / SolidWorks:** open any file from `step/`. It comes in as a solid body you can edit directly: Press/Pull, add holes, cut panel lines, re-fillet. STEP files don't carry the feature history.
- **Full parametric control:** `src/build_models.py` defines every vehicle from a list of cross-section stations (x, half width, top and bottom half heights, shape exponents) in full-scale metres, plus fin planforms. Change a number and run `pip install cadquery trimesh reportlab`, then `python make_all.py` to regenerate the STL, STEP and preview files, and `python make_manual.py` to rebuild this manual.
- **Changing scale:** pass `scale=` to a build function (for example `build_x43a(scale=16)`). Pins, slot clearances and the stand peg stay at printable sizes.

Troubleshooting

Problem	Fix
Halves rock or show a gap	Elephant foot on the split face: shave or sand it flat on sandpaper taped to glass.
Fin tab won't go in	Sand the tab faces, or regenerate with a larger CLEAR value.
Thin nose or edges curl	Lower nozzle temp 5 C, raise the minimum layer time, use the part-cooling fan at 100 %.
Duct roof (X-43A) sags	Bridging: slow bridge speed, fan 100 %. A little sag is hidden inside the duct.
Body lifts at the ends	Use a brim, clean the bed with IPA, raise the bed temp 5 C.
Peg loose in socket	One wrap of tape on the peg, or a drop of glue if you want it permanent.

Sources for the public figures quoted: NASA Armstrong fact sheets (X-43A, X-15), AFRL / Boeing X-51A releases, DARPA HTV-2 releases. Shapes are approximate; check against reference photographs if accuracy matters for your use.